Journal of Embedded System Security and Intelligent Systems
E-ISSN: 2722-273X; P-ISSN: 2745-925X
https://journal.unm.ac.id/index.php/JESSI

Automatic Floor Stain Detection with Image Processing: A Practical Comparison of OpenCV and RGB Grayscale Conversion

Achmad Zulfajri Syaharuddin[1*], Sri Suci Indasari[2], Nurhikmayana Janna[3], Andi Afdhal Hilmi[4]

[1,2,3,4,] *Department of Informatics and Computer Engineering, State Polytechnic of Ujung Pandang, Makassar*

Corresponding e-mail : ajisan@poliupg.ac.id

ARTICLE INFO

Keywords:

Dirt Detection, Grayscale Conversion, Image Processing, OpenCV, Thresholding

Article History

Received: September 15, 2025
Revised : November 03. 2025
Accepted : November 11, 2025

ABSTRACT

This research develops an image processing-based floor stain detection system using grayscale conversion and binary thresholding. Two conversion approaches are compared: (i) a simple RGB grayscale formula and (ii) a built-in OpenCV function. The system uses a fixed intensity threshold of T=80 and classifies a floor as "dirty" if the detected area exceeds 20% of the image. Experiments are conducted on three floor types (plain, dark, patterned), five stain types (coffee, oil, ink, plastic, chalk), and two lighting conditions. Results show that the performance of both methods is very close with an average difference of ≈0.07% and a maximum of 0.6%; the simple formula is suitable for resource-limited devices, while OpenCV is more robust to color/lighting variations. The main contributions of this paper are (1) a practical comparison of two grayscale conversion pathways for cleanliness monitoring, (2) a simple decision rule based on the percentage of dirty area that aligns with cleanliness perception, and (3) an analysis of implementation implications for real-time systems in cleaning robots/IoT. Future directions include adaptive thresholding and morphology integration to improve reliability in dynamic environments. (Replace the current abstract paragraph containing T=80 and the 20% rule with the version above. The 20% policy reference is already explained in your manuscript).

This is an open access article under the CC BY-SA license

To cite this article : Achmad Zulfajri Syaharuddin. (2025). Automatic Floor Stain Detection with Image Processing: A Practical Comparison of OpenCV and RGB Grayscale Conversion. Journal of Embedded System Security and Intelligent Systems, 6(4), 630-640. Doi. 10.59562/jessi.v6i4.9805

INTRODUCTION

In most cases, human traffic is heavy in crucial places such as households, healthcare facilities, restaurants, and industries. Therefore, the possibility of garbage, stains (footprint stains and other forms of stains), and debris is quite high, causing safety and hygiene problems. Cleaning is often considered a repetitive task, typically performed in less-than-ideal environments. Cleaning is an important task to maintain a standard of living.

Floor cleanliness is an important aspect in various environments, such as households, healthcare facilities, restaurants, and industry [1]. Clean floors not only reflect good aesthetics, but also play an important role in maintaining health and safety. However, the floor cleaning process often still relies on conventional methods that rely on human visual inspection, which can be inefficient and subjective. Several studies have been conducted to be able to detect waste to provide real-time visualization of cleanliness levels, and can detect dirty areas based on human traffic patterns and the presence of stains or garbage [2].

The advancement of digital image processing technology has enabled the automation of tasks such as dirt and stain detection. Image-based monitoring systems can provide real-time

information about cleanliness levels, allowing better decision-making in maintenance and cleaning operations. In previous research, room cleanliness detection was carried out based on digital images by comparing the Support Vector Machine (SVM) and Neural Network (NN) methods [3], some of which used sensors, robots [4], and deep learning [5]. One of the commonly used approaches in digital image analysis is the thresholding algorithm that allows the separation of objects from the background based on the difference in pixel intensity in an image. Thresholding is used to set the number of degrees of grayness present in the image [6]. This algorithm has been widely used from various sectors such as health to perform breast cancer image segmentation [7], disease in shallots [8], flower image segmentation [9], vehicle fire and smoke segmentation [10], Wireless Body Sensor Network [11], image encryption [12] and so on, by applying thresholding, the system can automatically identify stains or dirt on the floor surface, thus increasing the efficiency and effectiveness of the cleaning process.

In this study, an automatic floor stain detection system is developed using the thresholding algorithm. Two grayscale conversion methods are compared: (1) a simple RGB-based formula, and (2) the built-in grayscale conversion from the OpenCV library. Both methods are tested on several floor types (plain, dark, and patterned) and under different lighting conditions. The performance of each approach is evaluated based on its accuracy in detecting stains and its suitability for implementation on low-resource devices such as microcontrollers or IoT-based systems.

The main goal of this study is to determine a practical and efficient image-processing-based approach for detecting floor stains. The results are expected to contribute to the development of automated cleaning or monitoring systems that can support hygiene maintenance in both household and industrial environments.

METHOD

This research was conducted to design a stain or dirt detection system on the floor using a digital image processing technique based on a thresholding algorithm. The procedure begins with taking digital imagery using a camera to capture the condition of the floor surface. In this study, several scenarios will be carried out to be able to compare the accuracy of the Thresholding Algorithm, that is:

Table 1. Research Scenario

Floor Variations	Types of Stains
Plain Floor	Coffee
	Ink
	Oil
	Plastic
Dark Floor	Chalk
	Ink
	Oil
	Plastic
Patterned Floor	Coffee
	Ink
	Oil
	Plastic

The flow of the research conducted in this study is as follows.

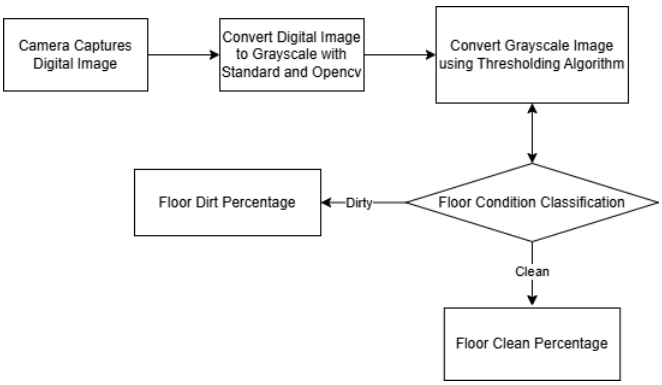


Figure 1. Research Flow

The digital imagery obtained is then converted into grayscale format to simplify the processing process and reduce color complexity. Grayscale is a very important technique for simplifying color imagery while still retaining important structural information. By converting the RGB color image to grayscale, we can reduce computational complexity and highlight important features for further analysis [13].

Once the image is in grayscale format, the next stage is to apply a thresholding algorithm to convert the grayscale image into a binary image (black and white). At this stage, a threshold of 80 is used from the range 0-255. This means that any pixel with an intensity value below 80 will be converted to black pixels (indicating possible blemishes), while pixels with values above that threshold will be white (indicating a clean area). In digital image processing, thresholding is one of the simplest methods to segment images. Thresholding is a common method of segmenting the region by dividing the image into several groups of pixels with a value that is lower, larger, or equal to the threshold value. Several thresholding techniques have been developed, such as the global method based on the gray histogram, the global method based on the local property, and adaptive thresholding [14].

Then, the system calculates the total number of pixels as well as the number of black pixels. If the number of black pixels is more than 20% of the total pixels, then the system concludes that the floor is in a dirty condition. Conversely, if the number of black pixels is less than or equal to 20%, then the system concludes that the floor is clean.

Data was obtained by taking various images of floor surfaces in different conditions, such as plain floors, dark floors, and patterned floors, where each condition was treated with a type of dirt on the floor. The test is carried out by processing each image through stages as described in the system flow. Each result was analyzed to measure the accuracy of the system in identifying stains, as well as to evaluate how sensitive the thresholding was to variations in lighting, floor type, and camera position.

In this study, a fixed threshold value of T=80 is applied to grayscale images to classify dirty pixels. While this method provides a simple and effective solution under controlled lighting and consistent floor conditions, it presents limitations when dealing with varying environmental factors. The fixed threshold may not account for significant differences in illumination, surface types, or the diversity of stains, which can result in false positives or negatives. For example, when

lighting conditions change (e.g., shadows or glare), the pixel intensity might fluctuate, leading to misclassification of clean areas as dirty or vice versa.

Adaptive thresholding offers a potential improvement by dynamically adjusting the threshold value based on local pixel intensity variations. Unlike global thresholding, which applies a single threshold value to the entire image, adaptive thresholding calculates the threshold for each pixel based on its local neighborhood. This approach allows the system to account for different lighting conditions and surface textures, improving the accuracy of stain detection. For instance, in areas with shadows, adaptive thresholding can lower the threshold to capture darker areas, while in bright spots, it can increase the threshold to avoid false detections.

digital images were captured using a standard camera with a resolution of 800x600 pixels, providing a total of 480,000 pixels per image. The camera used is a commonly available model suitable for capturing floor surfaces in everyday environments. The camera was positioned at a consistent height of 1 meter above the floor surface to maintain uniformity across all images. The images were taken under two distinct lighting conditions: natural daylight (for outdoor conditions) and artificial lighting (for indoor conditions) to simulate common cleaning environments.

To ensure the system could handle diverse lighting scenarios, the test environment included both shaded and well-lit areas. Shaded areas were chosen to simulate low-light conditions, while well-lit areas were used to represent normal or bright lighting. This setup helped assess how the system performed under varying illumination and whether the thresholding algorithm was sensitive to these changes. The resolution of 800x600 pixels was chosen to balance image clarity with processing speed, ensuring that the system could process images quickly without compromising detection accuracy.

System performance measurement was carried out by comparing system detection results using simple grayscale and OpenCV grayscale. The term "OpenCV" stands for "open source computer vision." Its architecture consists of pre-programmed software, databases, and plugins with support for integrating computer vision applications [15]. Evaluation is carried out based on the level of accuracy of stain detection and the resulting cleanliness value.

RESULTS AND DISCUSSION

This research aims to develop a detection system for stains or dirt on the floor surface based on digital imagery using a thresholding algorithm. The procedure starts from the acquisition of digital images by the camera, which functions as the main device in capturing the floor surface.

Image Acquisition and Pre-Processing

Digital images were taken using a camera with a resolution of 800 x 600, which is worth 480.000 pixels, RGB format. Total pixels are used as a divisor to calculate the level of dirt on the floor. Each image was taken under varying lighting conditions (shaded and unshadowed) on three types of floors, namely plain floors, dark floors, and patterned floors.

Once the image is obtained, it is converted to a grayscale image. A grayscale image is a digital image that has one channel value on each pixel. These values are used to indicate color intensity. Grayscale images are stored in 8-bit format for each pixel, allowing for as many as 256 intensities (0-255). The standard grayscale formula:

$$Grayscale\ (x, y): (R\ + G + B)/\ 3 \qquad (1)$$

Where R, G, and B are the red, green, and blue color components at pixel coordinates (x,y). Then, a simple grayscale comparison was carried out with the OpenCV grayscale using the formula [16]:

$$Grayscale\ (x, y): (0.114\ .\ B) + (0.59\ .G) + (0.299\ .R) \qquad (2)$$

OpenCV stores images in BGR format instead of RGB. This weight follows the Y' luminance standard of the Y'UV/Y'IQ color model used in analog and digital televisions. OpenCV is used for image preprocessing and conversion between color spaces to improve image quality, input, and output [17].

Thresholding and Segmentation

The grayscale image is then converted to a binary image using a thresholding algorithm. The threshold value is still used at 80 in the range of 0-255 value scale, with the following rules:

$$Pixel_{bin}(x, y) = \begin{cases} 0\ jika\ Gray(x, y) < T \\ 255\ jika\ Gray\ (x, y) \geq T \end{cases} \qquad (3)$$

Where Gray(x,y) is a grayscale image, image(x,y) is a binary image, and T is a threshold value. An illustration of the pixel change in the thresholding process with a value of 80 is shown in **Figure 2**.

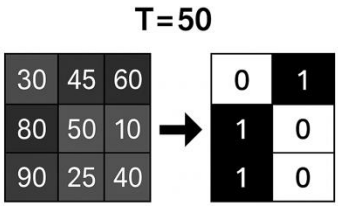


Figure 2. Illustration of changes in pixel values in the thresholding process

When the threshold value is set on the pixel data, the value below the threshold value will be 0, while the value above the threshold value will be 1.

Classification of Floor Conditions

In this study, the floor is classified as dirty if the area identified as dirty exceeds 20% of the total area of the segmented image, while it is classified as clean if the percentage is below that value. The determination of the 20% threshold is based on the visual perception of the cleanliness of the environment. Previous research in the field of visual ergonomics and hygiene perception states that humans begin to feel disturbed about the cleanliness of a surface when the dirty area has covered more than 15–30% of the field of view. [18] Although international hygiene standards such as ISO 14644 (for clean spaces) do not set numerical thresholds in the form of a percentage of dirty areas, the implementation of the 20% threshold is considered to be in line with the precautionary principles used in maintenance [5].

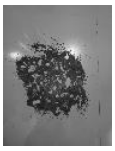
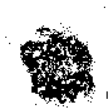
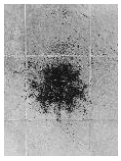
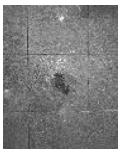
Testing and Data Collection

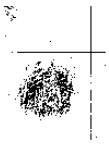
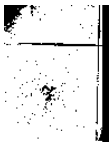
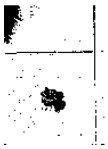
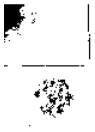
Testing was conducted on 12 floor image samples in three categories:

- Floor Variations,
- Plain floor,
- Dark floor

Each floor category has a different stain type, such as coffee, oil, plastic, ink, chalk, and is tested with a thresholding algorithm, and the detection results will compare the standard grayscale formula with the OpenCV grayscale based on the level of floor dirt. Based on the results of the tests carried out.

Table 2. Research Results and Testing

Types of Stains	Picture (sumPixel = 480.000)	Grayscale Result	Threshold Result (T=80)	Total Dirty Pixel (dirtyPixel)	Level of Impurity (dirtyPixel / sumPixel x 100)	Simple Formula Result (*)	OpenCV Result(**)
Clean Floor							
Coffee				2.181.719 px* 2.180.524 px**	17,8935 %* 17,8837 %**	Classification result was **17,89**% = clean floor	Classification result was **17,88**%, = clean floor
Ink				171.449 px* 179.772 px**	1,4062% * 1,47447 %**	Classification result was **1,4**% = clean floor	Classification result was **1,5**% = clean floor
Oil				27.361 px* 28.317 px**	0,2244% * 0,23224 %**	Classification result was **0,22**% = clean floor	Classification result was **0,23**% = clean floor
Plastic				436.800 px* 436.520 px**	3.5825% * 3.5802% **	Classification result was **3,58**% = clean floor	The floor classification result was **3,58** = clean floor
Dark Floor							
Chalk				1.193.054 px* 1.193.054 px**	31,5287 %* 31,3404 %**	Classification result was **31,53**% = dirty floor	Classification result was **31,53**% = dirty floor
Ink				562.509 px* 554.685 px**	4,6135% * 4,5493% **	Classification result was **4,61**% = clean floor	Classification result was **4,55**% = clean floor

Oil				491.465 px* 486.745 px**	4,0308% * 3,9921% **	Classification result was **4,03**% = clean floor	The classification result was **4**% = clean floor
Plastic				766.815 px* 764.771 px**	6,2891% * 6,2723% **	Classification result was **6,3**% = clean floor	Classification result was **6,27**% = clean floor
Patterned Floor							
Coffee				1.261.948 px* 1.262.344 px**	10,35%* 10,3532 %**	Classification result was **10,35**% = clean floor	Classification result was **10,35**% = clean floor
Ink				928.858 px* 933.730 px**	7,6181% * 7,6581% **	Classification result was **7,62**% = clean floor	Classification result was **7,66**% = clean floor
Oil				992.844 px* 992.934 px**	8,1429% * 8,1436% **	Classification result was **8,14**% = clean floor	Classification result was **8,14**% = clean floor
Plastic				808.296 px* 808.233 px**	6,6293% * 6,6288% **	Classification result was **6,63**% = clean floor	Classification result was **6,63**% = clean floor

The findings from this study suggest that the thresholding algorithm, when applied to grayscale images of floor surfaces, can effectively classify areas as "clean" or "dirty." The accuracy of this method, especially when using the OpenCV library for grayscale conversion, shows potential for real-world applications like autonomous floor cleaning robots. These robots could leverage the system to detect and navigate dirty areas autonomously, ensuring efficient cleaning in environments like hospitals, offices, or industrial facilities. The 20% dirt threshold is particularly valuable in this context, as it aligns with human perception of cleanliness, enabling the robots to prioritize high-traffic areas or spots that exceed cleanliness thresholds.

This study's results indicate that even with relatively simple hardware, the system can contribute significantly to operational efficiency. By detecting stains early and automating the cleaning process, businesses can reduce downtime, improve the accuracy of their cleaning routines, and lower labor costs. Moreover, the low computational cost of the system makes it feasible to implement on devices with limited processing power, such as microcontrollers used in autonomous robots or IoT sensors.

Another practical challenge is the need for consistent image quality. The system relies on high-quality images for accurate stain detection, but factors such as camera position, dirt on the camera lens, or low-quality sensors can lead to suboptimal results. Ensuring that the image acquisition setup is stable and well-maintained, along with using high-quality cameras and sensors, is essential for consistent performance. Moreover, in highly dynamic environments, where the floor conditions might change rapidly, maintaining real-time processing and adaptability becomes a critical factor for success.

CONCLUSIONS

Provide a statement that what is expected, as stated in the "Introduction" chapter can ultimately result in "Results and Discussion" chapter, so there is compatibility. Moreover, the prospect of the development of research results and application prospects of further studies can be added to the next section (based on the results and discussion).

This study compared two grayscale conversion methods for detecting floor dirtiness, focusing on the accuracy of pixel detection after the thresholding process. The two methods compared are: (1) the conversion method using a simple grayscale formula, and (2) the conversion method using the OpenCV2 library. For each method, the image processing process included the conversion of the color image to grayscale, followed by the thresholding process using the intensity value threshold (T = 80) with a pixel value of 80 out of 255 to produce a black-and-white binary image. Black pixels (value 0) are considered to represent a dirty area, while white pixels (value 1) represent a clean area.

The following is a comparison table of simple grayscale and grayscale OpenCV results on **Table 2**.

Table 2. Comparison of Stain Detection Results

Types of Stains	Simple Formula Result (*)	OpenCV Result (**)	Comparison of Results
Clean Floor			
Coffee	17.89%	17.88%	0.01%
Ink	1.40%	1.50%	0.10%
Oil	0.22%	0.23%	0.01%
Plastic	3.58%	3.58%	0.00%
Dark Floor			
Chalk	31.53%	31.53%	0.00%
Ink	4.61%	4.55%	0.60%
Oil	4.03%	4.00%	0.03%
Plastic	6.30%	6.27%	0.03%
Patterned Floor			
Coffee	10.35%	10.35%	0.00%

Ink	7.62%	7.66%	0.04%
Oil	8.14%	8.14%	0.00%
Plastic	6.63%	6.63%	0.00%

The results on **Table 2**. Shows that the total number of pixels in each image is fixed, i.e., 480,000 pixels (e.g., 800×600 resolution). This value is used as a divisor to calculate the level of impurity (%), which is formulated as follows:

$$Level\ of\ Impurity = \frac{Dirty\ Pixel\ Count}{Total\ Pixel\ Count}\ x\ 100 \qquad (4)$$

For example, for coffee stains on a plain floor, the manual grayscale method detects 2,181,719px dirt, resulting in a dirt rate of 17.89%, while the OpenCV method detects 2,180,524px dirt, with a dirt rate of 17.88%. This small difference shows that both methods have a very close performance of 0.01%. Likewise, ink stains in dark flow have a fairly large ratio of 0.6%. The rest of the study had almost the same value between the standard grayscale and OpenCV grayscale results.

In general, the results indicate that the thresholding approach can effectively distinguish between clean and dirty areas based on image intensity, offering a practical solution for cleanliness monitoring. This system has the potential to be integrated into IoT-based monitoring platforms or autonomous cleaning robots, contributing to more efficient and consistent hygiene maintenance in public or industrial spaces.

For future research, improvements can focus on implementing adaptive thresholding to handle varying illumination levels and integrating morphological filtering techniques to reduce noise and increase detection accuracy. In addition, expanding the dataset with different floor materials and stain types can further enhance the model's reliability for real-world applications.

AKNOWLEDGMENT (IF ANY)

The author would like to gratefully acknowledge the support of the Department of Informatics and Computer Engineering, Ujung Pandang State Polytechnic, for the support and facilities provided during the research. Special thanks are also extended to all colleagues and students who contributed during the image acquisition and testing phases. Additionally, appreciation is given to the editorial team of SiTekIn for the opportunity to publish this work.

REFERENCES

[1] Suprihatin, T. Susilowati, S. Muljani, L. Edahwati, and Sutiyono, "Teknologi Pembuatan Pembersih Lantai Sereh untuk Menambah Kesejahteraan Kelompok Tani Kosagrha Lestari Surabaya," *Jurnal Pengabdian Teknik Mesin*, 2023.

[2] B. Ramalingam, A. V. Le, Z. Lin, Z. Weng, R. E. Mohan, and S. Pookkuttath, "Optimal selective floor cleaning using deep learning algorithms and reconfigurable robot hTetro," *Sci Rep*, vol. 12, no. 1, Dec. 2022, doi: 10.1038/s41598-022-19249-7.

[3] S. Suparni, H. Rachmi, and A. Al Kaafi, "Detection of Room Cleanliness Based on Digital Image Processing using SVM and NN Algorithm," *SinkrOn*, vol. 7, no. 3, pp. 777–783, Jul. 2022, doi: 10.33395/sinkron.v7i3.11479.

[4] R. O. Wiyagi and K. Wada, "Toilet Floor Trash Detection System for Unidentified Solid Waste," *Journal of Robotics and Mechatronics*, vol. 36, no. 6, pp. 1558–1567, 2024, doi: 10.20965/jrm.2024.p1558.

[5] M. S. Vivekanandan and T. Jesudas, "Deep Learning Implemented Visualizing City Cleanliness Level by Garbage Detection," *Intelligent Automation and Soft Computing*, vol. 36, no. 2, pp. 1639–1652, 2023, doi: 10.32604/iasc.2023.032301.

[6] D. Lasmaria Simangunsong, "SEGMENTASI CITRA PADA CITRA ASLI BUAH JERUK BERDASARKAN NILAI THRESHOLDING," 2025. [Online]. Available: http://jurnal.goretanpena.com/index.php/JSSR

[7] K. Djunaidi, N. Lanang, A. Syahdu, and D. Kuswardani, "Penerapan Metode Thresholding : Entropy dan Mathematical Morphology Pada Segmentasi Citra USG Kanker Payudara," vol. 11, no. 2, 2022, doi: 10.33322/kilat.v11i2.1503.

[8] M. Gusti Tri Ananda, W. Pardosi, and A. Maysari Br Sembiring, "Literatur Review: Metode Thresholding Untuk Mengidentifikasi Penyakit Pada Daun Bawang Merah," 2024.

[9] P. William, S. 1}, Y. F. Riti, and C. K. Herijanto, "ANALISIS PERBANDINGAN ALGORITMA REGION GROWING DAN OTSU THRESHOLDING PADA SEGMENTASI CITRA BUNGA," vol. 18, pp. 19–27, 2023.

[10] M. Rezki, S. Nurdiani, R. Ade Safitri, M. Ifan Rifani Ihsan, M. Iqbal, and U. Nusa Mandiri JlJatiwaringin, "Segmentasi Api dan Asap Pada Kebakaran Dengan Metode K-Means Clustering," 2022. [Online]. Available: http://jurnal.bsi.ac.id/index.php/co-science

[11] S. K. Arumugam *et al.*, "A Novel Energy Efficient Threshold Based Algorithm for Wireless Body Sensor Network," *Energies (Basel)*, vol. 15, no. 16, Aug. 2022, doi: 10.3390/en15166095.

[12] Z. Hua, K. Zhang, Y. Li, and Y. Zhou, "Visually secure image encryption using adaptive-thresholding sparsification and parallel compressive sensing," *Signal Processing*, vol. 183, Jun. 2021, doi: 10.1016/j.sigpro.2021.107998.

[13] D. Aaisyah Salmaa Putri Atmaja and U. Kalimantan Timur, "Implementation of Grayscale Image Transformation and Histogram Equalization Methods in Digital Image Processing," *Sistem Kendali & Jaringan) E-ISSN*, vol. 4, pp. 2808–3520, 2025, doi: 10.58982/krisnadana.v4i2.739.

[14] S. Jardim, J. António, and C. Mora, "Image thresholding approaches for medical image segmentation-short literature review," in *Procedia Computer Science*, Elsevier B.V., 2023, pp. 1485–1492. doi: 10.1016/j.procs.2023.01.439.

[15] R. T. H. Hasan and A. B. Sallow, "Face Detection and Recognition Using OpenCV," *Journal of Soft Computing and Data Mining*, vol. 2, no. 2, pp. 86–97, Oct. 2021, doi: 10.30880/jscdm.2021.02.02.008.

[16] B. AlHumaidan, S. Alghofaily, M. Al Qhahtani, S. Oudah, and N. Nagy, "Parallel Image Processing: Taking Grayscale Conversion Using OpenMP as an Example," *Journal of Computer and Communications*, vol. 12, no. 02, pp. 1–10, 2024, doi: 10.4236/jcc.2024.122001.

[17] Y. Bhanu, P. Reddy, Y. Prem Kumar, U. Srihari, M. K. Saranya, and T. Kumanan, "Deep Learning: Powered Grayscale to Color Conversion Using Open CV," 2025.

[18] A. T. Hoang, V. V. Pham, and X. P. Nguyen, "Integrating renewable sources into energy system for smart city as a sagacious strategy towards clean and sustainable process," *J Clean Prod*, vol. 305, Jul. 2021, doi: 10.1016/j.jclepro.2021.127161.